

Akshita Santra

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EDUCATION

The University of Texas at Austin, Austin, TX

May 2028

Bachelor of Science in Computer Science, Minor in Business

- **Relevant Coursework:** Data Structures & Algorithms, Operating Systems, Linear Algebra, Computer Architecture

SKILLS

Programming Languages: Java, Python, C++, C, JavaScript, TypeScript, C#, Swift

Backend & Databases: FastAPI, Node.js, Express, REST APIs, PostgreSQL, MySQL, InfluxDB, MongoDB, Prisma, SQL, WebSockets

AI & Data: OpenAI API, AI Agents, MCP (Model Context Protocol), TensorFlow, PyTorch, scikit-learn, Pandas, NumPy, OpenCV

Tools & Platforms: Docker, AWS, Git, GitHub, React, Vite, Figma, MATLAB, R

Certifications: Oracle Certified Associate - Java SE 8 Programmer (*Issued May 2024*)

EXPERIENCE

IBM, *Software Developer Intern*, San Jose, CA

May 2026 - Present

- Developed an AI agent to enable engineers to analyze benchmark data across MySQL/InfluxDB with conversational prompts.
- Built a Python MCP server exposing 5+ read-only tools for data retrieval and analysis.
- Implemented a FastAPI backend integrating a natural-language query assistant, database connectors, and MCP tooling to support a standalone app.

Aramco Americas, *Software Engineering Intern*, Houston, TX

May 2025 - August 2025

- Programmed two robots that collect oil well data to enhance field safety and support more sustainable resource extraction, by developing modular C++ firmware for 20+ hardware components including motors.
- Developed Python-based UI/UX for both robots with BLE connectivity and multithreaded state workflows, leveraging NumPy/Pandas for data processing and Matplotlib dashboards for sensor visualization.
- Enabled reliable MATLAB/Arduino data exchange via Modbus Comms with zero packet loss.

NanoAssembly Lab, *Research Assistant*, Austin, TX

December 2024 - December 2025

- Systematized a 1D-CPNN machine learning framework for small-angle X-ray scattering (SAXS) analysis, reducing error rates by ~73%, helping to facilitate the design of more efficient drug-delivery nanoparticles.
- Developed and trained a deep learning model on 24,000+ simulated SAXS profiles in Python utilizing Tensorflow and Scikit-learn, achieving an $R^2=0.987$ and improving prediction accuracy by ~79% over baseline methods.
- Created a GUI and bundled executable with Python using tkinter, enabling researchers to run micelle structure predictions 10^3 - $10^6\times$ faster without coding expertise or specialized hardware.

Society of Women Engineers (SWE) - Visualizing Math, *Web Application Developer*, Austin, TX

August 2024 - May 2025

- Developed a full-stack educational website on Algebra and Geometry with quizzes and interactive graphing tools (HTML, CSS, JavaScript), with an average 30% improvement in quiz scores after interactive practice.
- Integrated Firebase Firestore in production and MongoDB + AWS in development, to track and retrieve user progress.

Microsoft, *Blacks at Microsoft Apprenticeship Program*, Houston, TX

July 2023 - August 2023

- Gained proficiency in Python and taught daily classes to 20+ apprentices, strengthening team programming skills.
- Directed a 1st-place hackathon team to build smart glasses that record visual experiences to aid dementia patients.

PROJECTS - Full Portfolio: <https://akshitasantra.github.io/>

Dayzy

December 2026

- Designed (Figma) and built a mutli-tab SwiftUI iOS app to log activities, visualize time usage, and generate summaries.
- Implemented a SQLite-backed persistence layer with custom SQL queries for real-time stats and activity recommendations.

Health IoT Dashboard

October 2025

- Developed a full-stack Healthcare IoT Dashboard with a responsive frontend (React + Vite) and robust backend (Node.js/Express + Prisma + PostgreSQL), enabling real-time patient telemetry with <1s latency via REST APIs + WebSockets.
- Automated build, containerization and deployment with Docker + Render, reducing setup time from 30 min to <3 min and ensuring consistent production environments.

LEADERSHIP & ACTIVITIES

Student Engineers Educating Kids, *VP Corporate*, Austin, TX

January 2025 - Present

- Lead and run weekly STEM mentoring sessions for 25+ elementary students, ensuring an engaging learning environment.
- Secured \$10,000+ in corporate sponsorship funding to support the org by cultivating relationships with industry partners.

FIRST Robotics Team, *Team Captain/Programming Lead*, Tomball, TX

August 2020 - May 2024

- Coded all functionality for the 2023 and 2024 robot in Java; successfully competed at the World Championships.